Development of the kidney

3- Metanephros:

- It is the permanent kidney.

- It begins to function at the 9th week.

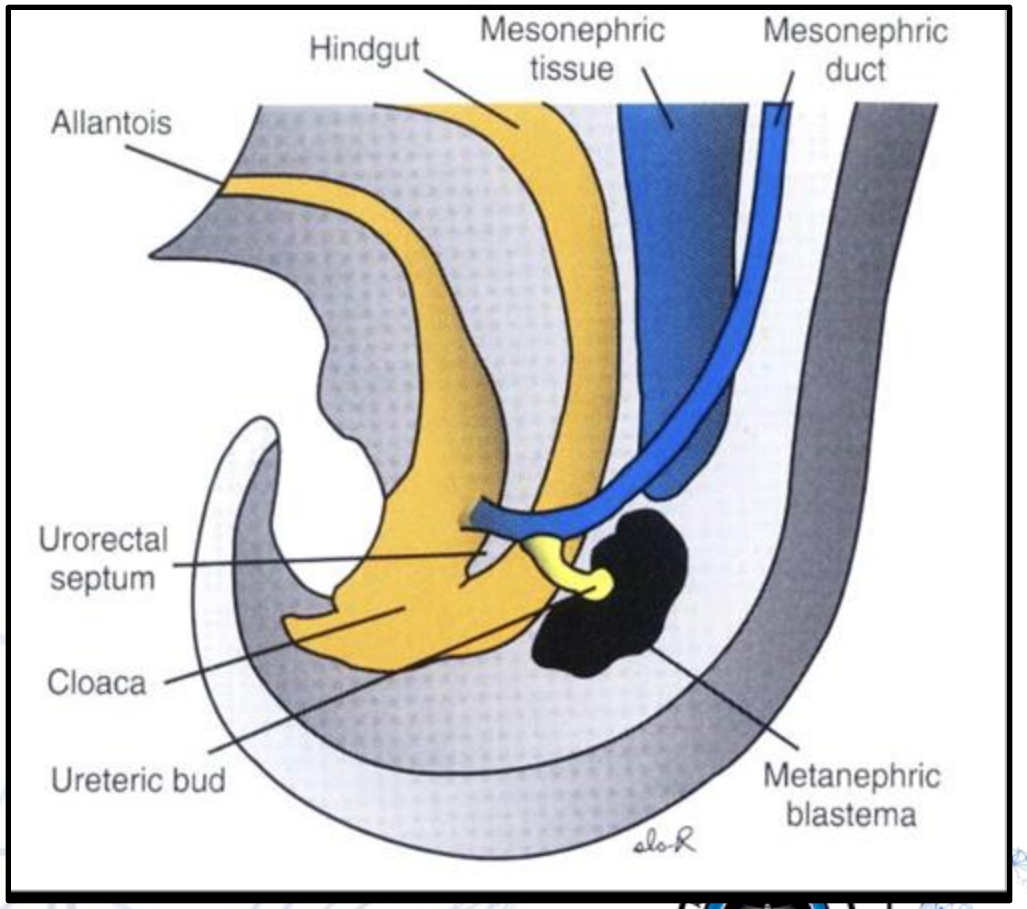
- Urine is produced and released into amniotic fluid.

Site: lower lumber and sacral region.

Structure: It develops from two parts;

Collecting part

Secretory part

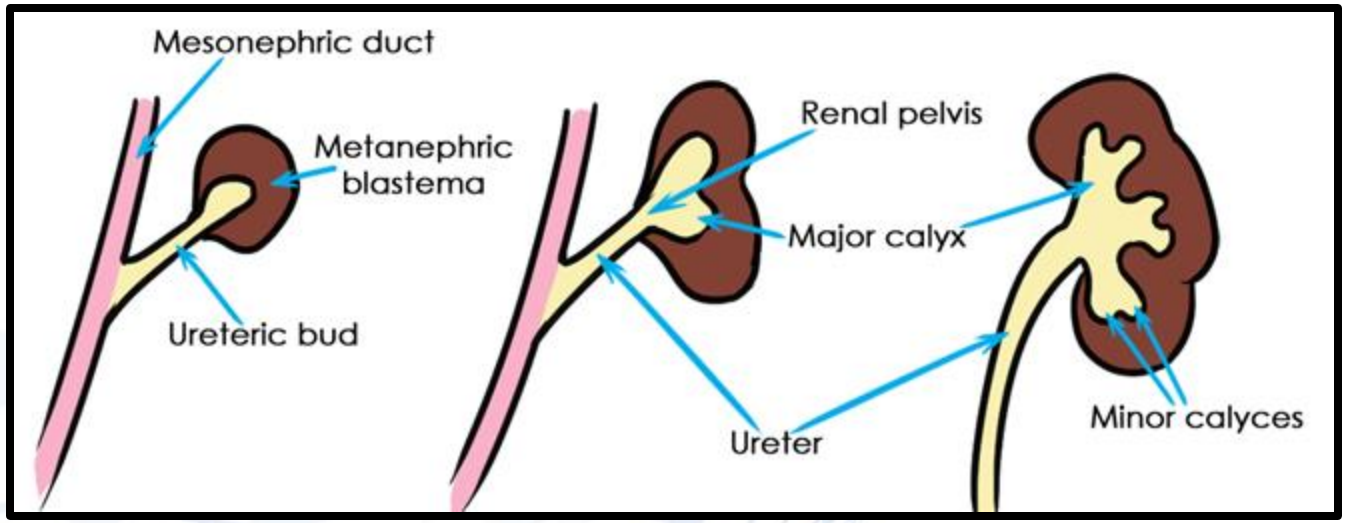


ANU MEDICAL SCHOOL PROGRAM

Development of the kidney
3- Metanephros

1- The collecting part:

- The ***ureteric bud*** arises as an outgrowth from the lower end of the mesonephric duct near the cloaca.
- The ureteric bud elongates forming the ureter, renal pelvis, major calyces, minor calyces, collecting ducts and tubules



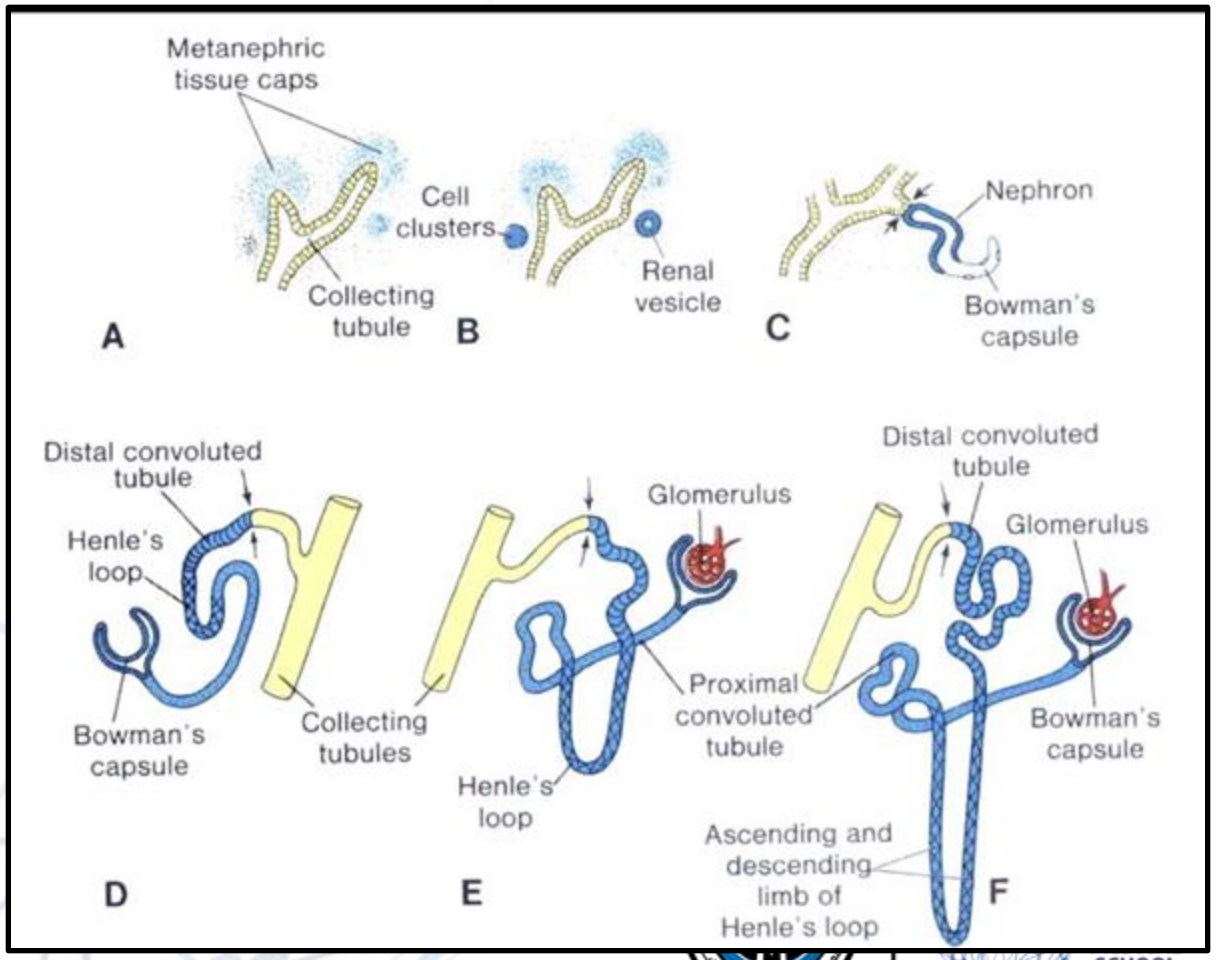
ANU MEDICAL SCHOOL PROGRAM

Development of the kidney
3- Metanephros

2- The secretory part:

- The ***nephrogenic cord*** (caudal part of the intermediate mass of mesoderm) is divided into segments called ***metanephrogenic cap***. Each segment lies opposite one collecting tubule.

- The metanephrogenic cap forms the ***nephron*** (glomerulus, Bowman's capsule, proximal and distal convoluted tubules and loop of Henle.



SCHOOL PROGRAM